

Linear Momentum

- 1) The momentum of a particle is given by $\vec{p}(t)$
 $\vec{p}(t) = 4 \cdot 8 t^2 \hat{i} - 8 \cdot 0 \hat{j} - 8 \cdot 9 t \hat{k}$. What is the force as a function of time?

Newton's second law $\Rightarrow \vec{F}(t) = \frac{d\vec{p}(t)}{dt}$

$\vec{F}(t) \rightarrow$ force as a function of time

$\vec{p}(t) \rightarrow$ momentum as a function of time

$$\vec{p}(t) = (4 \cdot 8 t^2) \hat{i} - 8 \cdot 0 \hat{j} - (8 \cdot 9 t) \hat{k}$$

$$\vec{F}(t) = \frac{d\vec{p}(t)}{dt} = \frac{d}{dt} (4 \cdot 8 t^2 \hat{i} - 8 \cdot 0 \hat{j} - 8 \cdot 9 t \hat{k})$$

$$= 4 \cdot 8 \times 2 t \hat{i} - 0 \hat{j} - 8 \cdot 9 \hat{k}$$

$$\vec{F}(t) = ((9 \cdot 6 t) \hat{i} - 8 \cdot 9 \hat{k}) \text{ N}$$

2. a) $m = 0.045 \text{ Kg}$ $v_i = 0 \text{ m/s}$ (ball at rest)

$$v_f = 45 \text{ m/s}$$

$$\text{Impulse } J = \Delta p = m \cdot \Delta v$$

$$= m (v_f - v_i)$$

$$= 0.045 (45 - 0)$$

$$= 0.045 \times 45$$

$$= 2.025 \text{ N}\cdot\text{s}$$

b)

~~the~~

$$J = F_{\text{avg}} \cdot \Delta t$$

$$F_{\text{avg}} = \frac{J}{\Delta t}$$

$$\Delta t = 3.5 \text{ ms} = 3.5 \times 10^{-3} \text{ s}$$

$$= \frac{2.025}{3.5 \times 10^{-3}} = \frac{2025}{3.5} = 578.57 \text{ N}$$

$$(1996 - 1416) \approx 580 \text{ N}$$

3)

mass of car $m_1 = 1000 \text{ kg}$ mass of truck $m_2 = 3000 \text{ kg}$ velocity of car $v_1 = 30 \text{ m/s}$ east v_2 of truck $= -20 \text{ m/s}$ west

a) conservation of momentum

$$p_i = p_f$$

$$p_i = m_1 v_1 + m_2 v_2$$

$$= 1000(30) - 3000(20)$$

$$= 30000 - 60000$$

$$= -30000 \text{ kg m/s}$$

$$v_f = \frac{p_i}{m_1 + m_2} = \frac{-30000}{1000 + 3000} = \frac{-30}{4} = -7.5 \text{ m/s}$$

$$\Rightarrow 7.5 \text{ m/s west}$$

$$b) \quad V_1' = \frac{(m_1 - m_2) V_1 + 2 m_2 V_2}{m_1 + m_2}$$

$$V_2' = \frac{(m_2 - m_1) V_2 + 2 m_1 V_1}{m_1 + m_2}$$

car's final velocity

$$\Rightarrow V_1' = \frac{(m_1 - m_2) V_1 + 2 m_2 V_2}{m_1 + m_2}$$

$$= \frac{(1000 - 3000) 30 + 2(3000)(-20)}{1000 + 3000}$$

$$= \frac{-60000 - 120000}{4000} = \frac{-180000}{4000}$$

$$= -\frac{90}{2} = -45 \text{ m/s}$$

Car goes west² at 45 m/s

truck's final velocity

$$\Rightarrow V_2' = \frac{(m_2 - m_1) V_2 + 2 m_1 V_1}{m_1 + m_2}$$

$$= \frac{(3000 - 1000)(-20) + 2(1000)(30)}{1000 + 3000}$$

$$= \frac{-40000 + 60000}{4000}$$

$$= \frac{20000}{4000} = 5 \text{ m/s}$$

Truck goes east at 5 m/s

$$4) \quad m_1 = 6 \text{ kg} \quad \vec{v}_{1i} = (2 \text{ m/s}) \hat{i} \quad m_2 = 3 \text{ kg} \quad \vec{v}_{2i} = (-5 \text{ m/s}) \hat{j}$$

+ x-axis - y-axis

$$\vec{P}_i = \vec{P}_f$$

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}_f$$

$$\vec{P}_{1i} = m_1 \vec{v}_{1i} = 6 (2) \hat{i} = 12 \text{ kg m/s } \hat{i}$$

$$\vec{P}_{2i} = m_2 \vec{v}_{2i} = 3 (-5) \hat{j} = -15 \text{ kg m/s } \hat{j}$$

$$\vec{P}_i = \vec{P}_{1i} + \vec{P}_{2i} = 12 \hat{i} - 15 \hat{j} \text{ kg m/s}$$

$$\vec{v}_f = \frac{\vec{P}_i}{m_1 + m_2} = \frac{12 \hat{i} - 15 \hat{j}}{6 + 3} = \frac{12}{9} \hat{i} - \frac{15}{9} \hat{j}$$

$$= \frac{4}{3} \hat{i} - \frac{5}{3} \hat{j}$$

$$= (1.33 \hat{i} - 1.67 \hat{j}) \text{ m/s}$$

magnitude of final velocity

$$\Rightarrow \sqrt{(1.33)^2 + (-1.67)^2}$$

$$\Rightarrow 2.14 \text{ m/s}$$

5) $m_A = m_B = m$ $\vec{V}_{Ai} = 40 \text{ m/s}$ $\vec{V}_{Bi} = 0$
 $\theta_A = 30^\circ$ above x-axis $\theta_B = 45^\circ$ below x-axis

(a) $\vec{P}_i = \vec{P}_f$
 $m\vec{V}_{Ai} + m\vec{V}_{Bi} = m\vec{V}_{Af} + m\vec{V}_{Bf}$

$\therefore m_A = m_B = m$ $\vec{V}_{Bi} = 0$
 (m gets cancelled)
 $\vec{V}_{Af} + \vec{V}_{Bf} = \vec{V}_{Ai}$

x-component

$$\vec{V}_{Ai} \cdot x = V_{Af} \cdot x + V_{Bf} \cdot y$$

$$40 = V_{Af} \cos 30^\circ + V_{Bf} \cos 45^\circ$$

y-component

$$\vec{V}_{Ai} \cdot y = V_{Af} \cdot y + V_{Bf} \cdot y$$

$$0 = V_{Af} \sin 30^\circ - V_{Bf} \sin 45^\circ$$

-ve because

$$V_{Af} \sin 30^\circ = V_{Bf} \sin 45^\circ$$

$$V_{Af} = V_{Bf} \frac{\sin 45^\circ}{\sin 30^\circ}$$

$$= V_{Bf} 1.414$$

y-component
 is -ve as
 $\theta_B = 45^\circ$ is below
 axis

$$40 = V_{Bf} 1.414 \cos 30^\circ + V_{Bf} \cos 45^\circ$$

$$40 = V_{Bf} (1.414 \cos 30^\circ + \cos 45^\circ)$$

$$V_{Bf} = \frac{40}{1.932} = 20.7 \text{ m/s}$$

$$V_{Af} = 1.414 V_{Bf} = 1.414 \times 20.7 = 29.3 \text{ m/s}$$

Speed of asteroid A after collision is 29.3 m/s
 Speed of asteroid B is 20.7 m/s

b) Fraction of original KE of A that dissipates

$$= \frac{KE_{\text{dissipated}}}{KE_{Ai}}$$

$$= \frac{\frac{1}{2} m V_{Ai}^2 - \left(\frac{1}{2} m V_{Af}^2 + \frac{1}{2} m V_{Bf}^2 \right)}{\frac{1}{2} m V_{Ai}^2}$$

$$= \frac{V_{Ai}^2 - (V_{Af}^2 + V_{Bf}^2)}{V_{Ai}^2} = 1 - \frac{V_{Af}^2 + V_{Bf}^2}{V_{Ai}^2}$$

$$V_{Ai}^2 = 40^2 = 1600 \text{ (m/s)}^2$$

$$V_{Af}^2 = (29.3)^2 = 858.5 \text{ m/s}^2$$

$$V_{Bf}^2 = (20.7)^2 = 428.5 \text{ (m/s)}^2$$

$$\text{fraction} = 1 - \frac{858.5 + 428.5}{1600}$$

$$= 1 - \frac{1287}{1600}$$

$$= 1 - 0.804$$

$$= 0.196 = 19.6\%$$

The fraction of the original kinetic energy of asteroid 1 that dissipates is approximately 0.196 (19.6 %). This means the collision is inelastic.

$$6) \quad \frac{1}{2} mv^2 = mg \Delta y$$

$$v = \sqrt{2g \Delta y}$$

$$v = \frac{2mv \cos \phi}{2m} = v \cos \phi$$

$$\cancel{v}^2 = 2gh$$

$$v^2 = 2gh$$

$$h = \frac{v^2 \cos^2 \phi}{2g}$$

$$h = \Delta y \cos^2 \phi$$

$$\Delta y = R/2$$

$$\cos^2 \phi = 1/2$$

$$h = R/2 \times 1/2 = R/4$$

$$\boxed{h = R/4}$$